

DS

Assignment-1

Aim: To write a C program to perform basic matrix operations such as addition, subtraction, multiplication and transpose of a matrix using 2D arrays.

Algorithm:

Step 1: START

Step 2: Declare 2 matrices (arrays) • A, B, C variables
 $r_1, c_1, r_2, c_2, i, j, k$

Step 3: Display Menu

- i) Addition
- ii) Subtraction
- iii) Multiplication
- iv) Transpose
- v) Exit

Step 4: Input choice

Step 5: Use switch case

Case 1 - Input r_1, c_1 & i, j ; if $A \& B$ $C[i][j] = A[i][j] + B[i][j]$ & display C.

Case 2 - Input r_1, c_1 & elements of matrices A & B. Perform $C[i][j] = A[i][j] - B[i][j]$ & display C.

Case 3 - Input r_1, c_1, r_2, c_2 & elements of A & B. If $c_1 = r_2$, perform matrix multiplication & store result in C. Display C.

Otherwise display "Multiplication not possible".

Case 4: Input r_1, c_1 & elements of A. Find transpose using $C[i][j] = A[j][i]$ & Display C.

Case 5: Exit the program. Display Invalid choice
Step 6: STOP

FAQs

Q1. Explain matrix addition, subtraction, multiplication, transpose with suitable examples & state the conditions required for each operation.

Ans. Addition and Subtraction:

Condition: Both matrices must have the same dimensions.

Operation: Element-wise addition/subtraction.

EX -

$$\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} + \begin{bmatrix} 3 & 3 \\ 4 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ 6 & 6 \end{bmatrix}$$

Multiplication:

Condition: Columns of Matrix A must equal rows of Matrix B.

Operation: Dot product of rows of A and columns of B.

EX - $(1 \ 2) \times \begin{bmatrix} 3 \\ 4 \end{bmatrix} = 1(3) + 2(4) = 11$

Transpose:

Condition: None

Operation: Swap rows with columns.

EX -

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

Objectives:-

- To understand the concept of 2-dimensional arrays in C.
- To perform arithmetic operations on matrices.
- To implement matrix addition and subtraction.
- To implement matrix multiplication using nested loops.
- To find the transpose of a matrix.
- To develop programming skills using arrays and loops.

Theory-

A matrix is a rectangular arrangement of elements organized in rows and columns. In C programming, matrices are represented using two-dimensional arrays. Matrix operations are widely used in computer graphics, image processing, machine learning, and engineering applications.

1. Matrix Addition

Condition: Both matrices must have the same number of rows and columns.

Formula:

$$C[i][j] = A[i][j] + B[i][j]$$

2. Matrix Subtraction

Condition: Both matrices must have the same dimensions.

Formula: $C[i][j] = A[i][j] - B[i][j]$

3. Matrix multiplication.

Condition: Number of columns of the first matrix be equal to the number of rows of the second matrix.

Formula:

$$C[i][j] = \sum [A[i][k] \times B[k][j]] \text{ where } k = 0$$

4. Transpose of a matrix

The Transpose of a matrix is obtained by interchanging its rows and columns.

Formula: $A^T[i][j] = A[j][i]$

These operations are implemented using 2-D arrays, nested loops, and conditional statements.

Step 6: Display

Step 7: End

Step 8: End

Step 9: End

Step 10: End

Step 11: End

Step 12: End

Step 13: End

Step 14: End

Step 15: End

Step 16: End

Step 17: End

Step 18: End

Q2. Describe how 2D arrays are used to represent matrices in C programming.

A matrix is represented in C using a 2D array. Each row of the matrix corresponds to one row of the array, and each column corresponds to one column of the array.

```
int A [3][3];
```

This creates a 3x3 matrix.

Input Using Nested loops

```
for (i=0; i<3; i++)
```

```
{
```

```
    for (j=0; j<3; j++)
```

```
    {
```

```
        scanf ("%d", &A[i][j]);
```

```
    }
```

```
}
```

Display Using Nested loops

```
for (i=0; i<3; i++)
```

```
{
```

```
    for (j=0; j<3; j++)
```

```
    {
```

```
        printf ("%d", A[i][j]);
```

```
    }
```

```
    printf ("\n");
```

```
}
```

manipulating Elements

To increase every element by 3:

```
for (i=0; i<3; i++)  
{  
    for (j=0; j<3; j++)  
    {  
        A[i][j] = A[i][j] + 3;  
    }  
}
```

Q3. Write the algorithm for performing matrix multiplication and explain why compatibility between matrices necessary.

Ans. Step 1 - START

Step 2 - Read rows and columns of matrices A and B.

Step 3 - Check compatibility

Step 4 - Input elements of matrices A and B.

Step 5 - Multiply matrices using nested loops.

Step 6 - Display matrix C.

Step 7 = STOP

Compatibility is necessary cause matrix multiplication is possible only when columns of first matrix = Rows of Second

Q4. Discuss the applications of matrix operations in real world domains.

Ans. Matrix operations are widely used in computer graphics, Image Processing, machine learning

1. Computer graphics: Rotation
 - Scaling
 - Translation
 - 3D animation

• Used to transform images and objects efficiently.

2. Image Processing: Image enhancement
 - Filtering
 - Compression
 - Edge detection

3. Machine learning - Neural networks
 - Deep learning
 - Linear regression

4. Scientific Computing - Weather forecasting
 - Space research
 - Physics simulations
 - Numerical methods

5. Explain the role of nested loops in implementing matrix operations in C. Illustrate how nested loops are used for addition, multiplication and transpose of matrices.

Ans. Nested loops are used to access each element systematically.

Outer loop $\rightarrow$ Rows
Inner loop $\rightarrow$ Columns

Matrix Addition -

```
for (i=0; i < R; i++)  
{  
    for (j=0; j < C; j++)  
    {  
        C[i][j] = A[i][j] + B[i][j];  
    }  
}
```

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Step - Step

Discuss the applications of matrix operations in
graphics, Image Processing, machine learning